Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cylinders and spheres, and volume. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	5
Topic	Three dimensional shapes II
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume
Theme	—
Curriculum objectives	• identify properties of cylinders and spheres • find the volume of a cube, a cuboid and a prism
Previous knowledge	• properties of cubes and cuboid • properties of pyramids and cones
Lesson sequence	• identify properties of cylinders and spheres — the teacher displays shapes of cylinders and spheres; the student identify the properties; the teacher draws the shapes on the board and adds to the properties already mentioned by the students. Set for the students: 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces. • find the volume of a cube, a cuboid and a prism — the teacher explains how to derive the formula for volume of cube and cuboid, and works the volume of a prism as area of base times height; the student listen and ask questions where they do not understand; the student copy worked examples into their notes
Instructional materials	• shapes of cylinders and spheres.
Evaluation	• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high. • Calculate the volume of cuboid with dimensions 5m by 2m by 3m. • What is the volume of a cube of 4cm × 4cm × 4cm. • A rectangular tank hold 6m long by 2m wide holds $36m^3$ of water. How deep is the water in the tank.

Lesson objectives

1.	identify properties of cylinders and spheres
2.	find the volume of a cube, a cuboid and a prism

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cylinders and spheres, and volume

Class: JSS1 **Subject:** Mathematics **Week:** 5

Section 1: Properties of Cylinders and Spheres

Two common three-dimensional shapes are the cylinder and the sphere. You can tell each one apart by the faces, edges and vertices it has.

The cylinder

Look at the cylinder in fig-01. Think of a tin or a drum. A cylinder has a top and a bottom that are both circular, and a curved wall joining them.

Its properties are:

1. It has two circular faces and one curved face.
2. It has no vertex (no pointed corner).
3. It has two curved edges — the two circles where the curved face meets each circular face.
4. It has three faces altogether.

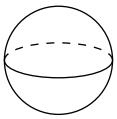
The sphere

Look at the sphere in fig-02. Think of a ball. Its whole surface is curved, with no flat parts at all.

Its properties are:

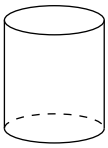
1. It is symmetrical from all directions — however you turn it, it looks the same.
2. It has no edge and no vertex.
3. It has one curved surface.

Visual



A sphere drawn as a circle with an ellipse across its middle to show the curved surface.

Visual



A cylinder drawn upright: an ellipse for the top face, two vertical sides falling from its ends, and a curve closing the base.

Worked example

Describe the faces, edges and vertices of a cylinder.

A cylinder has two circular faces and one curved face, so it has three faces altogether. Where each circular face meets the curved face, there is a curved edge, so it has two curved edges in total. Because a cylinder has no pointed corner, it has no vertex.

Practice

1. How many circular faces does a solid cylinder have?
2. A cylinder has how many faces altogether?
3. State how many edges and how many vertices a sphere has.

Section 2: Finding the Volume of a Cube, Cuboid and Prism

Volume measures how much space a solid takes up. For the solids you meet here, volume is found the same way every time: multiply the area of the base by the height. The only difference is how you work out that base area.

A cuboid is a box with rectangular faces. Its base is a rectangle, so:

Volume of a cuboid = area of base × height = length × breadth × height = $l \times b \times h$.

A cube is a special cuboid where length, breadth and height are all equal. If the side is l, then:

Volume of a cube = $l \times l \times l = l^3$.

Worked example — finding the volume of a cuboid
Calculate the volume of a cuboid of length 3 cm, breadth 2 cm and height 8 cm.

Volume of cuboid = $l \times b \times h$
length = 3 cm, breadth = 2 cm, height = 8 cm
Volume = 3 cm × 2 cm × 8 cm = 48 cm^3.

Worked example — finding the height of a cuboid
The volume of a cuboid is 24 cm^3. Its base is 6 cm by 2 cm. Find its height.

Volume = $l \times b \times h$
24 cm^3 = 6 cm × 2 cm × h
24 cm^3 = 12 cm^2 × h
Divide both sides by 12 cm^2:
24 cm^3 ÷ 12 cm^2 = h
h = 2 cm.

Worked example — finding the volume of a cube
Calculate the volume of a cube of side 3 cm.

Volume of cube = l^3
V = $(3\text{ cm})^3$ = 3 cm × 3 cm × 3 cm = 27 cm^3.

Worked example — finding the volume of a prism
A prism has two identical ends. The volume of a prism is still:

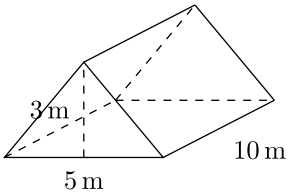
Volume of prism = area of base × height.

Consider a triangular prism (fig-03) whose triangular end has a base of 5 m and a height of 3 m, running back 10 m.

First find the area of the triangular end (the base of the prism):
Area of triangle = ½ × b × h = ½ × 5 × 3 = 7.5 m^2.

Then multiply by the length of the prism:
Volume of prism = 7.5 m^2 × 10 m = 75 m^3.

Visual



A triangular prism lying on its rectangular face. The triangular end has a base of 5 m and a height of 3 m, and the prism runs back 10 m.

Practice

1. Calculate the volume of a cuboid that is 8 cm long, 4 cm wide and 3 cm high.
2. Calculate the volume of a cuboid with dimensions 5 m by 2 m by 3 m.
3. What is the volume of a cube of side 4 cm?
4. A rectangular tank is 6 m long and 2 m wide, and it holds 36 m^3 of water. How deep is the water in the tank?

Check your work

Attempt every question before you read this page.

Properties of Cylinders and Spheres

1. 2

   A solid cylinder has a circular top and a circular bottom, so it has 2 circular faces.
2. 3

   The cylinder has two circular faces and one curved face, so it has $2 + 1 = 3$ faces altogether.
3. A sphere has no edge and no vertex.

   The whole surface of a sphere is one curved surface with no flat parts, so there are no edges where flat faces meet and no pointed vertices.

Finding the Volume of a Cube, Cuboid and Prism

1. 96 cm^3

   Volume = l × b × h = 8 cm × 4 cm × 3 cm = 96 cm^3.
2. 30 m^3

   Volume = l × b × h = 5 m × 2 m × 3 m = 30 m^3.
3. 64 cm^3

   Volume of cube = l^3 = 4 cm × 4 cm × 4 cm = 64 cm^3.
4. 3 m

   Volume = l × b × h, so 36 m^3 = 6 m × 2 m × h = 12 m^2 × h. Divide both sides by 12 m^2: h = 36 m^3 ÷ 12 m^2 = 3 m.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-2be3a8bf	**Date**	
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____